

ORIGINAL RESEARCH

Challenges and solutions of working under threshold supply-voltage, for CNTFET-based SRAM-bitcell

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Abstract

Recently, several studies were done on SRAM bitcells at different supply-voltages; upper, near or lower to threshold voltage. To the best of the author's knowledge, none of them discussed at threshold supply-voltage with proper subthreshold operations and Nano/Pico power-dissipations, hence this paper decides to investigate challenges and solutions of designing at $V_{DD} = V_{th}$, because this voltage will lead to having lower power consumptions. This research applies power-gating technique to adjust V_{DD} on V_{th} , and also utilises output-inverter to set Logic 1 at V_{DD} . Although 'power-gating' and 'output-inverter' were used in other works, this study renders specific points about them. In fact, the ability of power-gating technique in adjusting V_{DD} on V_{th} without any instabilities in bitcell operation was not reported before, as well as, the idea of employing output-inverter in read path for setting Logic-1 at V_{DD} . It also proposed SNM% as a useful figure-of-merit. The goal of this study is not to present new circuits for bitcell, but is to investigate challenges and solutions of working under ' $V_{DD} = V_{th}$ ' which these solutions can lead to having an optimum bitcell.

KEYWORDS

CNTFET, leakage current, power gating, SNM, SRAM, static and dynamic powers, subthreshold, threshold

1 | INTRODUCTION

Static Random Access Memory (SRAM) is one of the most important blocks in digital circuits that occupies a large area in chips. SRAMs are mainly used in computers and system-on-chips (SOCs) as cache memories which are recognised as static power consumption sources. The first Bipolar and MOS SRAMs were invented in 1963 by Robert Norman and in 1964 by John Schmidt, respectively, both at Fairchild Semiconductor [1, 2]. By scaling down of transistors over the years, supply and threshold voltages have been decreased simultaneously, but this V_{th} reduction leads to an exponential increase in leakage current (see S. R. Prasad et al in 2012 [3]). Hence, even though operating at low power supply achieves low dynamic power in write and read modes, SRAM leakage current is still a major issue due to its destructive impact on static (leakage) power in standby (hold) mode. Generally, stability and power consumption are the two main aspects of a SRAM-bitcell, and Static Noise Margin (SNM) is a figure-of-merit to determine its

performance stability and reliability. Dependency of read SNM on V_{DD} , and possibility of chips performance due to supply-voltage scaling in low-operating circuits such as subthreshold SRAMs were reported in Ref. [4].

After synthesising helical microtubules of graphitic carbon at NEC by Japanese researcher Sumio Iijima in 1991 [5] and presenting Carbon Nanotubes (CNTs) independently at NEC by Iijima et al [6] and at IBM by Bethune et al [7] in 1993, Carbon Nano Tube Field Effect Transistor (CNTFET), which uses CNT as the channel material instead of bulk silicon in the Metal Oxide Semiconductor Field Effect Transistor (MOSFET) were demonstrated in 1998 [8, 9]. In recent decades, many efforts have been made by researchers to offer optimum designs for bitcell. They declared that thanks to the CNTs, CNTFETs have been considered as a promising alternative for Complementary Metal Oxide Semiconductors (CMOSs) in low power SRAMs [10–75], however, CNFET fabrication process still has some imperfections, including the presence of metallic carbon nanotubes (m -CNTs), imperfect m -CNT removal processes, chirality

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drift, CNT doping variations, and density fluctuations, etc. [76–79]. Some of them exhibited comparative works to show the advantages of CNTFETs over CMOSs and FinFETs in bitcells due to extensive electrical properties of CNTs and excellent characteristics of CNTFETs [42–61]. They also reported that bitcells suffered from leakage power [10, 12, 36, 66, 67], Delay [36, 66–69], worse Read SNM [34, 49, 52, 66], negative impacts of nearthreshold supply voltage on delays and SNM [41, 68], subthreshold leakage currents [57], etc., and suggested different techniques, such as ‘decoupled R/W BLs’ [34–47, 66, 68] and ‘power-gating’ [41, 42, 63–65], to solve those problems.

In this respect, the improved CNTFET proposed in my previous work [62] with a channel length of 32 nm, diameter of 1.49 nm, chiral vector of (19, 0), pitch of 10 nm, and 3 nanotubes in 32 nm technology is utilised herein. These set-points were achieved based on a trade-off between delay and power, instead of PDP (Power-Delay Product), because the analysis in Ref. [62] revealed that using PDP causes a significant error in finding optimum adjustments for CNTFET. For example, the D-Latch achievements in Ref. [62] show 14 times reduction in power dissipation due to the set-points selected by trade-off factor, compared to those selected by PDP.

This paper tries to investigate the challenges and solutions of working under ‘ $V_{DD} = V_{th}$ ’ which these solutions can propose an optimum bitcell. Hence, first of all, it regenerates the conventional 6T SRAM-bitcell at threshold and near-threshold supply-voltages to present challenges ahead. Then 8T bitcell is analysed to reduce static and dynamic powers, and also to illustrate unreported feature of power-gating technique. Finally, a reliable 11T bitcell with ‘separate Read and Write Bit-Lines (R/W BLs)’ and ‘an output inverter in read path’ is suggested as a solution for Logic-1 level-reduction in read mode under ‘ $V_{DD} = V_{th}$ ’. This 11T bitcell has a subthreshold stable operation. Although several studies employed ‘decoupled R/W BLs’ and ‘power-gating’ techniques, none of them referred to this paper’s point. If it seems that using 11 Transistors causes some foot-print problems, you should note that previous works utilised more than 11 Transistors without any problems [12, 13, 15, 19, 20, 23, 28, 33, 58, 68]. The following analysis and simulations are done by HSpice, CNTFET Stanford Model, and Monte-Carlo Method.

2 | DESIGN AND ANALYSIS

As pictured in Figure 1, a typical SRAM memory is an array of the standard 6-transistor bitcells, which are selected by the address decoder and are written/read by the BL control unit. Each 6T bitcell is composed of two inverters ($X_{3,5}$, $X_{4,6}$) and two access switches (X_1 , X_2), as shown in Figure 2. In write mode, after driving BL and BLB by complementary voltage-levels as complementary data ([1,0] or [0,1] via the BL control unit), the word line (WL) is set high afterwards by the address decoder which leads to turn on the access switches X_1 and X_2 , and also to storage complementary data at the nodes Q and QN. In hold mode, the access switches are off (WL = 0) and the nodes Q and QN store their own data via internal feedbacks between the two inverters.

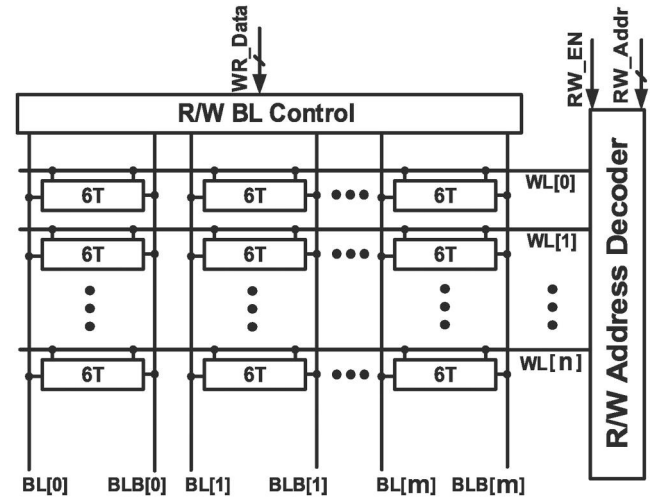


FIGURE 1 Structure of a typical Static Random Access Memory (SRAM) memory

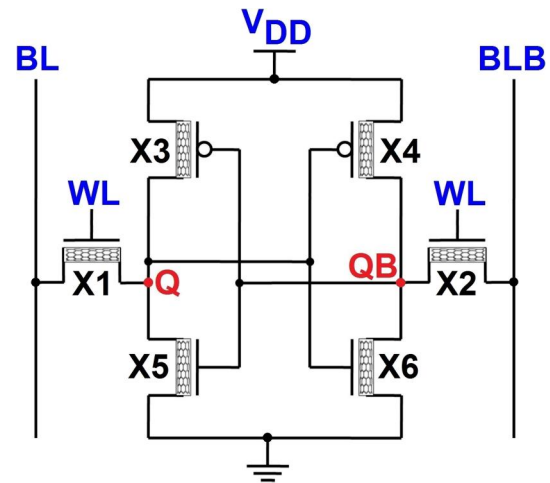


FIGURE 2 Structure of the standard 6T Static Random Access Memory (SRAM) bitcell based on Carbon Nano Tube Field Effect Transistor (CNTFET)

For example, assume $Q = 1$ and $QN = 0$ that in this case, Q turns-off X_4 and turns-on X_6 hence QN remains 0 which turns-off X_5 and turns-on X_3 to keep Q at 1, consequently data storage occurs in this bitcell. Finally, in read mode, WL becomes 1 again and the access switches are turned on to transfer data from ‘Q and QN’ to ‘BL and BLB’, respectively, because the BL control unit just senses the two bit-lines (BL and BLB) to receive the data in this mode.

Albeit this 6T structure is conventional for typical bitcells, it suffers from leakage power during standby mode [10], worse Read SNM [34], negative impacts of nearthreshold supply voltage on R/W delays and SNM and yield [41], subthreshold leakage currents [57], etc. By applying the reported CNTFET in Ref. [62], this 6T SRAM is regenerated herein under 0.3 and 0.33 V to specify related challenges. As discussed in Ref. [54], a CNTFET with 1.49 nm diameter has 0.29 V (≈ 0.3 V) threshold voltage (V_{th}). So this 6T bitcell is investigated at threshold (0.3 V) and nearthreshold (0.33 V) supply-voltages,

respectively. Observing Figure 3 reveals significant undershoots in the start-up current (-500 nA, -1μ A) during each WL's rise-time under 0.3 V supply (Figure 3e), because CNTFETs are sinking-to/sourcing-from the power supply, therefore V_{DD} as Logic-1 voltage-level in read mode drops sharply below V_{th} , and X_3 – X_6 cannot turn on/off perfectly (Figures 3c and d), so it causes a failure in the bitcell operation. We do not have this problem in write and hold modes, because, the Logic-1 voltage-level of entering data in write mode is previously set to V_{DD} by peripherals such as BL Control unit, and there is no start-up current in hold mode.

But in Figure 4e, despite of the larger overshoots and undershoots ($\pm 2\mu$ A) during WL's rise-time under 0.33 V supply voltage, the Logic-1 voltage-level never drops below V_{th} in read mode (Figure 4c,d), thus all CNTFETs can act properly. In this condition, Q and QN store correct values in each write process (Figure 4c), then BL and BLB sense these values in each read process (Figure 4d), however, this 6T bitcell still has some

problems in 'read mode', 'leakage current' and 'power dissipations' that has already been reported in [10, 12, 34, 36, 41, 49, 52, 57]. As observed, the current spikes in Figure 4e cause voltage-reduction (discharge) during the read process in Figure 4d. In fact, these spikes cause a kind of distortion in data pulse in read mode (Figure 4d). Thus 'discharges (distortion) in read mode', 'large overshoots/undershoots (spikes)' and 'drop of Logic-1 voltage-level in read mode' are the challenges ahead.

Table 1 shows that operating at nearthreshold V_{DD} (0.33 V) offers better 'power-dissipations' and 'stability (SNM %)', compared to other works with higher V_{DD} . Note that SNM% can be a proper figure-of-merit (FOM) for comparing the stability of bitcells which work at different V_{DD} . SNM% selects a minimum SNM (between 'SNM_{write}', SNM_{hold} and SNM_{read}') and calculates SNM_{min} per its V_{DD} , in percentage.

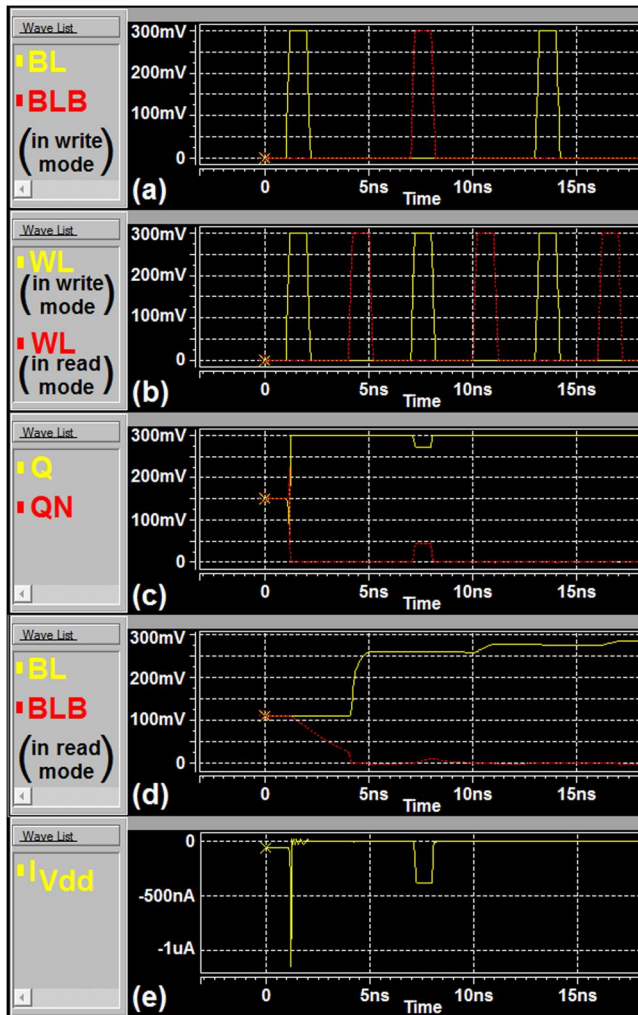


FIGURE 3 Simulation results for the standard 6T SRAM-bitcell under 0.3 V supply voltage: (a) data in write mode, (b) access switches in write/read modes, (c) values at nodes Q/QN, (d) data in read mode, and (e) current of power supply. Because of $V_{DD} = V_{th}$, the voltage-level of Logic-1 drops in read mode due to the microcurrent spikes, hence, the transistors cannot turn on/off perfectly

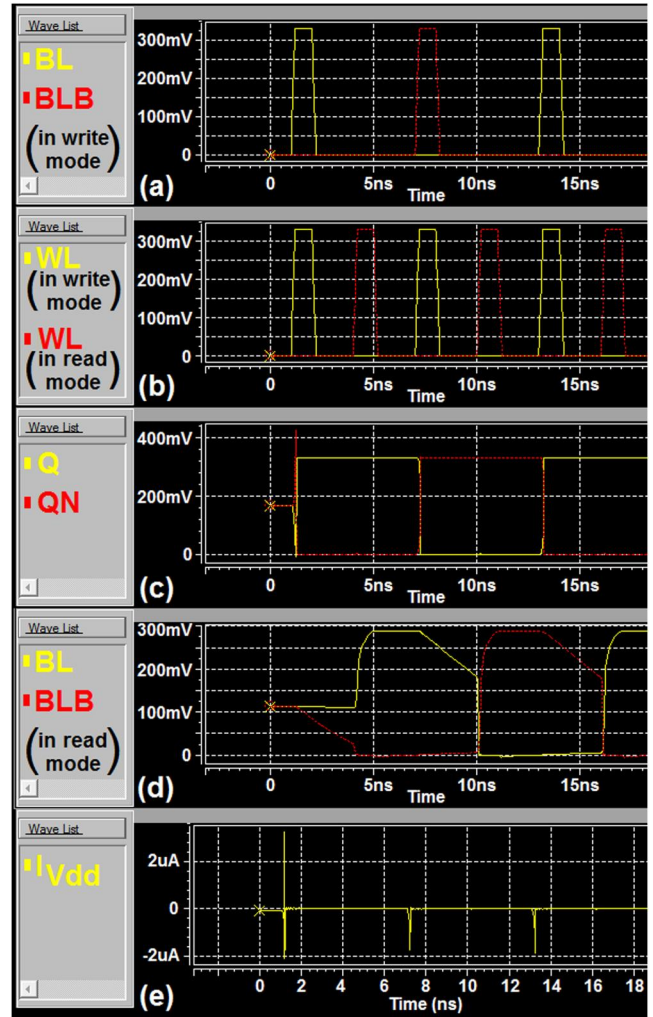


FIGURE 4 Simulation results for the standard CNTFET-based 6T SRAM-bitcell under 0.33 V supply-voltage: (a) data in write mode, (b) access switches in write/read modes, (c) values at nodes Q/QN, (d) data in read mode, and (e) current of power supply. The bitcell is written and read consecutively at nearthreshold supply-voltage. In write mode, the BL and BLB are driven as (a), the yellow word line (WL) turns on the access switches X_1 and X_2 as (b), and the nodes Q and QN store data as (c). Then, in read mode, the red WL turns on the access switches X_1 and X_2 as (b), and the BL and BLB read the data from the nodes Q/QN as (d). As observed in write mode (c), the data is a complete pulse, but in read mode (d), it is distorted. It still has microcurrent spikes (e)

TABLE 1 Performance comparison with the state-of-the-art for standard 6T 8T SRAMs

Work	Technology	Leakage current (A)	Static power (W)	Dynamic power (W)	Write SNM (mV)	Hold SNM (mV)	Read SNM (mV)	SNM (mV)	V _{DD} (V)	$\frac{SNM_{min}}{V_{DD}} \times 100 = SNM\%$
This work	32 nm CNTFET	272.6 p	89.9 p	9.29 n	143.1	140.1	147.3	---	0.33	$\frac{140.1}{330} \times 100 = 42\%$
[54]	32 nm CNTFET	---	---	13 n	---	---	---	338	0.9	$\frac{338}{900} \times 100 = 37\%$
[34]	32 nm CNTFET	205.5 p	185 p	---	---	---	---	290.8	0.9	$\frac{290.8}{900} \times 100 = 32\%$
[55]	32 nm CNTFET	6.24 n	6.24 n	4.42 μ	---	---	---	198	1	$\frac{198}{1000} \times 100 = 19\%$
[48]	32 nm CNTFET	---	---	---	---	354	155	---	0.9	$\frac{155}{900} \times 100 = 17\%$
[43]	32 nm CNTFET	---	---	---	312	284	118	---	0.9	$\frac{118}{900} \times 100 = 13\%$
[57]	32 nm CNTFET	---	---	---	123	---	198	---	0.9	$\frac{123}{900} \times 100 = 13\%$
[21]	32 nm CNTFET	---	---	8.75 μ	---	---	---	---	0.9	---

Indeed, considering ‘ratio’ instead of ‘SNMs with different V_{DD} ’, assists us to compare the works with different V_{DD} , see Table 1. Often proposing and testing proper figures-of-merit (FOMs) are very helpful in electronics [62, 80–82], for example, David Bol et. al. suggested Minimum-Energy Point for sub-45 subthreshold logic circuits in 2010 [83], hence the author suggests SNM% in this study.

By employing sleep transistors to implement power-gating methodology [63], the 6T bitcell is converted to an improved 8T one in Figure 5 which has small overshoots and undershoots (± 50 nA) with low power dissipations at ‘ $V_{DD} = V_{th} = 0.3V$ ’. Power-gating uses P-channel/N-channel transistors (called sleep transistors) as header-switches to shut off power-supplies to parts of a design in standby or sleep mode. As shown in Figure 6b, the ‘On’ signal is exerted to the sleep transistors X_7 and X_8 after WL’s rise time and before its fall time that it has a great impact on power consumptions due to limited-access of this bitcell to V_{DD} (Table 2). The technique shrinks current spikes (Figure 6e) and leakage currents (Table 2), hence power-gating makes it possible to working at threshold supply-voltage (0.3 V) instead of the nearthreshold one (0.33 V) while Q and QN have acceptable subthreshold operations (Figure 6c). This feature was not reported previously for power-gating technique [41, 42, 63–65]. As stated, M. Moradinasab and M. Fathipour in 2019 [57] reported an exponential increase in leakage current while Q and QN have subthreshold operations, but this 8T bitcell succeeds to achieve a low leakage current due to this feature of power-gating technique (Table 2). Also the bitcell renders better power-

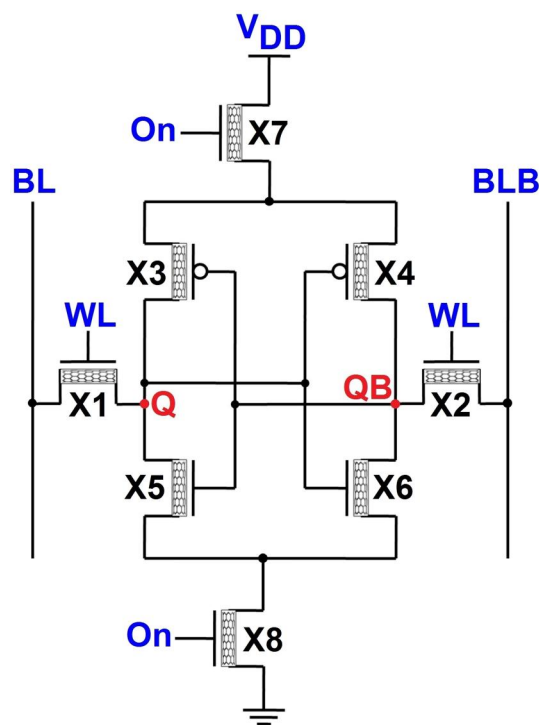


FIGURE 5 Structure of the power-gating 8T Static Random Access Memory (SRAM) bitcell based on Carbon Nano Tube Field Effect Transistor (CNTFET)

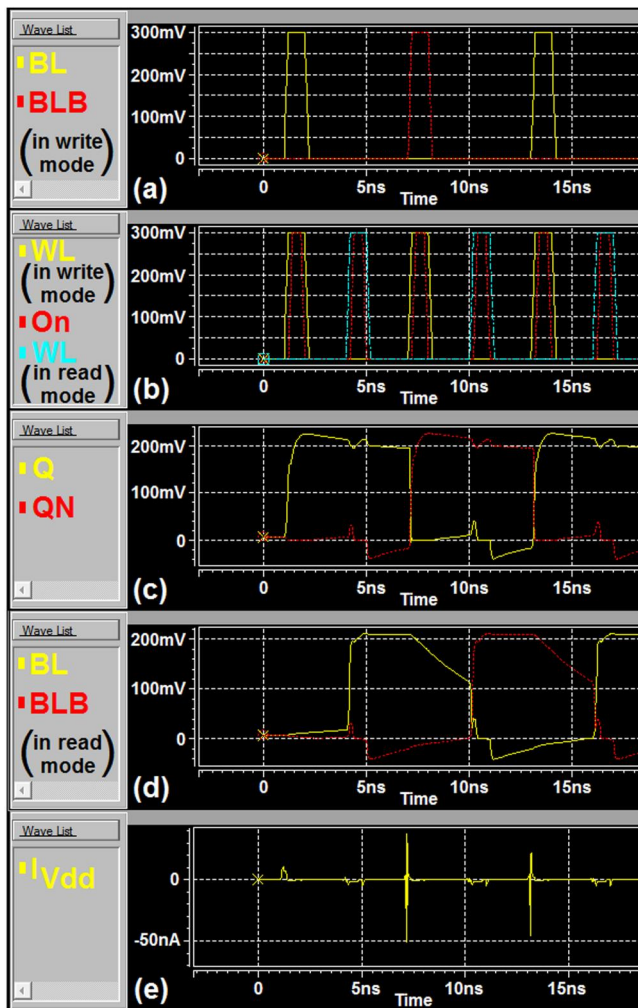


FIGURE 6 Simulation results for the CNTFET-based power-gating 8T SRAM-bitcell under 0.3 V supply voltage: (a) data in write mode, (b) access switches in write/read modes, (c) subthreshold values at nodes Q/QN, (d) data in read mode, and (e) current of power supply. The bitcell is written and read consecutively at threshold supply-voltage. In write mode, the BL and BLB are driven as (a), the yellow word line (WL) turns on the access switches X_1 and X_2 as (b), the red 'On' signal becomes 1 after yellow WL's rise-time until its fall-time as (b), and the nodes Q and QN store data as (c). Then, in read mode, the blue WL turns on the access switches X_1 and X_2 as (b), again the red 'On' signal becomes 1 after blue WL's rise-time until its fall-time as (b), and the BL and BLB read the data from the nodes Q/QN as (d). As observed in write mode (c), the data is a pulse, but in read mode (d), it is distorted. This 8 T bitcell has nanocurrent spikes (e), which is much better than the microcurrent spikes of 6T bitcell

dissipations and SNM%, compared to others (Table 2). Albeit it succeeds to achieve small spikes (Figure 6e) and solve 'large overshoots/undershoots (spikes)' (compare Figure 6e with 3e and 4e) and exhibit better characteristics than others, it cannot make any solutions for 'drop of Logic-1 voltage-level in read mode' and 'read-mode discharges (distortions)', hence we have to find an optimum bitcell.

To eliminate the effect of discharges in Figures 4d and 6d, read and write Bit-Lines (R/W Bit-Lines) must be separated as Figure 7. Read Bit-Line (RBL) senses data via 'inverter X_9 and X_{10} ' and 'read access-switch X_{11} ', while the write access-

TABLE 2 Performance comparison with the state-of-the-art for standard 8t srām bitcells

Work	Technology	Leakage Current (A)	Static Power (W)	Dynamic Power (nW)	Write SNM (mV)	Hold SNM (mV)	Read SNM (mV)	SNM (mV)	V_{DD} (V)	$\frac{SNM_{min}}{V_{DD}} \times 100 = SNM\%$
This work	32 nm CNTFET	54.82 p	16.4 p	0.31	156.3	130.5	156.4	-----	0.3	$\frac{130.5}{300} \times 100 = 43\%$
[41]	32 nm CNTFET	960 p	384 p	-----	-----	-----	-----	166.6	0.4	$\frac{166.6}{400} \times 100 = 41\%$
[40]	32 nm CNTFET	433.3 p	130 p	-----	-----	-----	-----	119.5	0.3	$\frac{119.5}{300} \times 100 = 39\%$
[34]	32 nm CNTFET	122.2 p	110 p	-----	-----	-----	-----	327.4	0.9	$\frac{327.4}{900} \times 100 = 36\%$
[55]	32 nm CNTFET	6.34 n	6.34 n	228	-----	-----	-----	354	1	$\frac{354}{1000} \times 100 = 35\%$

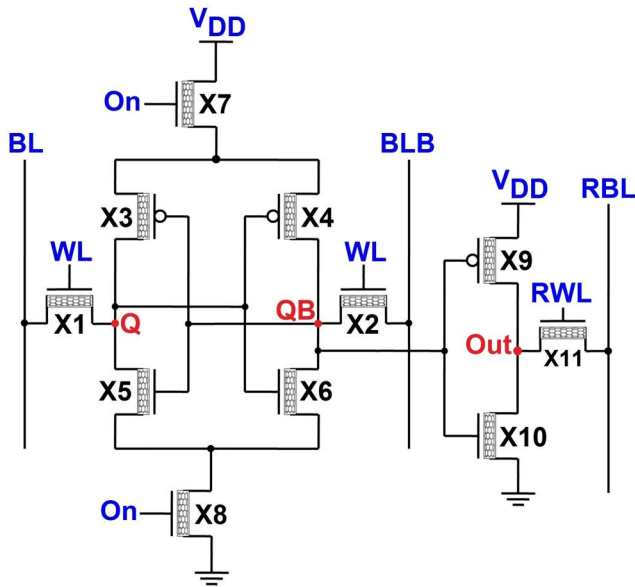


FIGURE 7 Structure of the CNTFET-based power-gating 11T Static Random Access Memory (SRAM) bitcell with decoupled R/W BLs that its read access-switch is after the inverter $X_{9,10}$. BL and BLB are for write performance, and Read Bit-Line (RBL) is for read performance

switches X_1 and X_2 are off. Figure 8 shows that even though the waveforms of Q, QN and RBL in this 11 T bitcell are improved, the spikes have growth to above -400 nA (Figure 8e) which is worse than the 8T one (-50 nA). In fact, because the gate of $X_{9,10}$ is directly impressed by ' X_1 and X_2 switching', it affects V_{DD} . Moreover, because the 11T one has subthreshold operation at nodes Q and QN, the voltage level of Logic-1 in RBL is below V_{DD} and V_{th} , which is not suitable for sensing by next devices or peripherals. Note that other CNTFET works employed similar decoupled R/W BLs circuits without X_9 or X_{10} , did not face these problems because of using ' $V_{DD} > V_{th}$ ' in their designs [34–47]. This 11T bitcell (Figure 7) just solves 'read-mode discharges (distortions)', but it increases the 'overshoots/undershoots (spikes)' and cannot solve 'the drop of Logic-1 voltage-level in read mode'.

As a solution, the read access-switch and the read inverter are replaced with each other in Figure 9 that it causes Logics 1 and 0 match V_{DD} and Gnd, respectively, (Figure 10d) while the bitcell has its proper subthreshold operation at nodes Q and QN (Figure 10c) without causing any instabilities. Furthermore, its nanocurrent spikes are as small as the 8T one (-50 nA), see Figure 10e. Obviously, this 11T bitcell succeed to solve challenges: 'drop of Logic-1 voltage-level in read mode', 'read-mode discharges (distortions)' and 'sharp overshoots/undershoots (spikes)'. Table 3 displays the achievements. Note that the write delay of 8T and 11T in Table 3 are the same because their write circuits are the same, but their read paths are different thus their read delays are various. In addition to the fine subthreshold performance (Figure 10c) and small spikes (Figure 10d), this 11T bitcell exhibits high stability (%SNM) compared to all 6T and 8T bitcells in Tables 1 and 2.

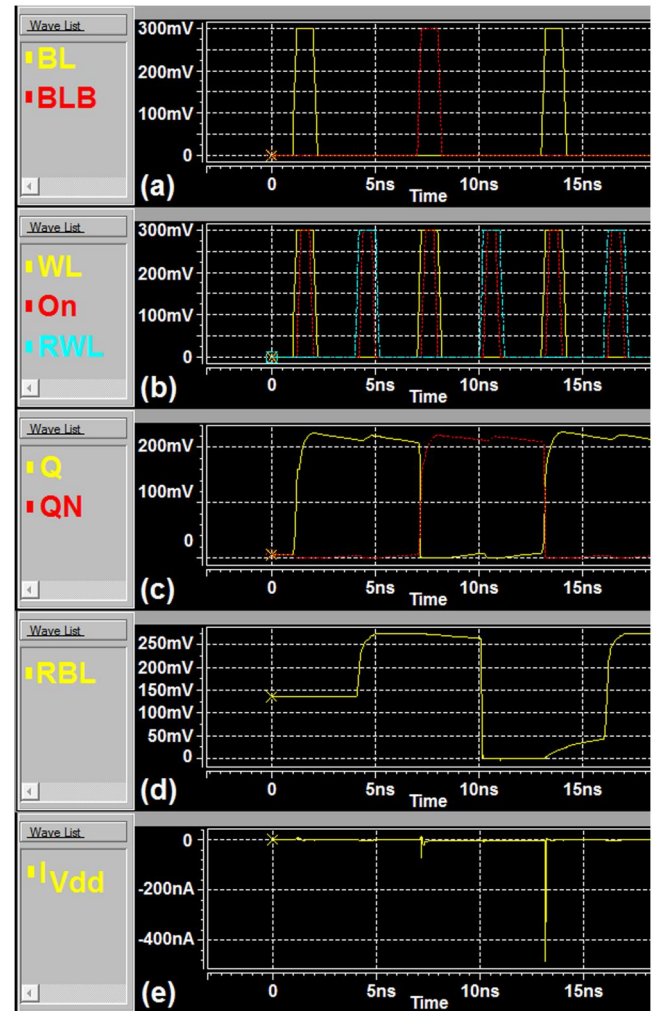


FIGURE 8 Simulation results for the CNTFET-based power-gating 11T SRAM-bitcell with decoupled R/W BLs under 0.3 V supply voltage that its read access-switch is after the inverter $X_{9,10}$: (a) data in write mode, (b) access switches in write/read modes, (c) subthreshold values at nodes Q/QN, (d) data in read mode, and (e) current of power supply. The bitcell is written and read consecutively at threshold supply-voltage. In write mode, the BL and BLB are driven as (a), the yellow word line (WL) turns on the access switches X_1 and X_2 as (b), the red 'On' signal becomes 1 after yellow WL's rise-time until its fall-time as (b), and the nodes Q and QN store data as (c). Then, in read mode, the blue WL turns on the access switches X_1 and X_2 as (b), again the red 'On' signal becomes 1 after blue RWL's rise-time until its fall-time as (b), and the Read Bit-Line (RBL) reads the data from the nodes Q/QN as (d). As observed, the data is a pulse in both write and read modes (c) and (d), without any distortion. This 11T bitcell has nanocurrent spikes (e), but it is larger than the nanocurrent spikes of 8T bitcell. Among the challenges, 'read-mode discharges (distortions)' is solved but 'drop of Logic-1 voltage-level in read mode' and 'large overshoots/undershoots (spikes)' are still unsolved

Also, it has an acceptable leakage current and static/dynamic powers (Table 3). Note that other works were not designed for operating at threshold supply-voltage ($V_{DD} = V_{th}$), as well as, they just reduced their delays at the cost of higher power-consumptions [10–75] which are not suitable for implementing in chipset of portable devices. As a result, this 11T bitcell (Figure 9) can be considered as a final optimum design that this paper tries to reach it from the beginning. Generally, this study

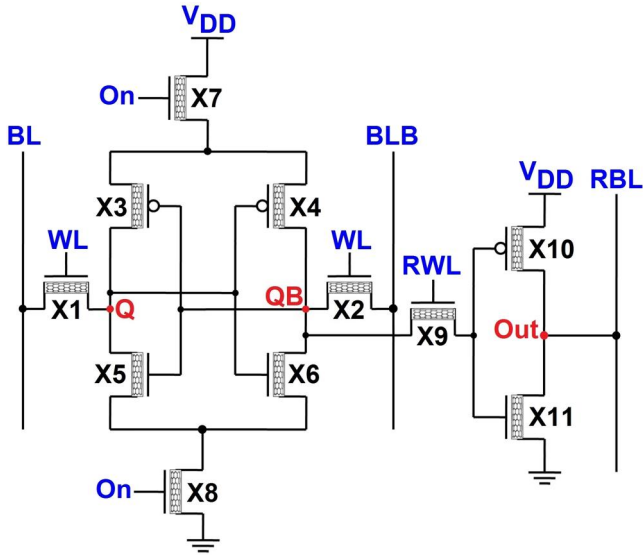


FIGURE 9 Structure of the CNTFET-based power-gating 11T Static Random Access Memory (SRAM) bitcell with decoupled R/W BLs that its read access-switch is before the inverter $X_{10,11}$. BL and BLB are for write performance, and Read Bit-Line (RBL) is for read performance

makes an attempt to discuss the challenges, solutions, and unreported ability of power-gating.

Albeit this work is not experimental, it applies different methods such as ‘Regenerating (the Standard 6T bitcell)’ and ‘Monte Carlo’ to demonstrate the reliability of this research. Actually, Monte Carlo is a computational method to estimate outputs based on input random samplings, that it is used herein (Figures 11–14) for predicting the function of the 6T, 8T and 11T bitcells in Figures 2,5,9. Note that, as discussed in Ref. [54], all CNTFET with 1.49 nm diameter have 0.29 V (≈ 0.3 V) threshold voltage (V_{th}), so they are not completely ‘ON’ at $V_{DD} < V_{th}$ and the bitcell behavior for $V_{DD} < 0.3$ V is not practical in Figures 11–14. The correct range is $V_{DD} > 0.33$ V for 6T and $V_{DD} > 0.3$ V for 8T and 11T.

The changes of V_{data} in read mode over V_{DD} are plotted in Figure 11, respectively, for the 6T, 8T and 11T bitcells, which shows ‘ $V_{data} = V_{DD}$ ’ only for the 11T bitcell. The two others have voltage reduction ($V_{data} < V_{DD}$) in read mode. Although the 11T bitcell has better V_{data} , its leakage-current and powers are higher than the 8T one (Figures 12–14).

In fact, the challenges ‘read-mode discharges (distortions)’, ‘large overshoots/undershoots (spikes)’ and ‘drop of Logic-1 voltage-level in read mode’ are solved in this 11T bitcell at the cost of higher power than the 8T one, but its static and dynamic powers are still smaller than ‘all 6T bitcells in Table 1’ and ‘other 8T bitcells in Table 2’, hence it can be an optimum 11T bitcell. Additionally, it has a high stability (% SNM) compared to all 6T and 8T bitcells in Tables 1 to 3. As stated, this 11T bitcell is able to render a fine function in write and read modes at $V_{DD} = V_{th}$ while Q and QN nodes have a proper subthreshold performance which is not reported before [10–75]. This subthreshold function does not lead to the exponential-increase in leakage-current reported in Ref. [57].

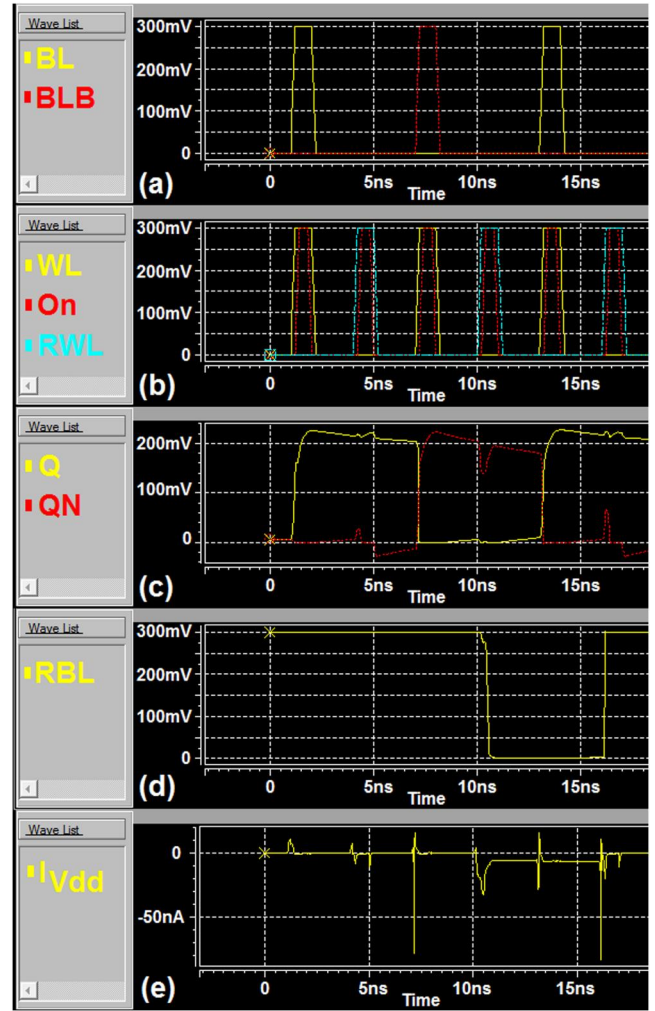


FIGURE 10 Simulation results for the CNTFET-based power-gating 11T SRAM-bitcell with decoupled R/W BLs under 0.3 V supply voltage that its read access-switch is before the inverter $X_{10,11}$: (a) data in write mode, (b) access switches in write/read modes, (c) subthreshold values at nodes Q/QN, (d) data in read mode, and (e) current of power supply. The bitcell is written and read consecutively at threshold supply-voltage. In write mode, the BL and BLB are driven as (a), the yellow word line (WL) turns on the access switches X_1 and X_2 as (b), the red ‘On’ signal becomes 1 after yellow WL’s rise-time until its fall-time as (b), and the nodes Q and QN store data as (c). Then, in read mode, the blue WL turns on the access switches X_1 and X_2 as (b), again the red ‘On’ signal becomes 1 after blue RWL’s rise-time until its fall-time as (b), and the Read Bit-Line (RBL) reads the data from the nodes Q/QN as (d). As observed, the data is a complete pulse in both write and read modes as (c) and (d), without any distortion. This 11 T bitcell has small nanocurrent spikes (e), similar to the nanocurrent spikes of 8T bitcell. The challenges ‘read-mode discharges (distortions)’, ‘drop of Logic-1 voltage-level in read mode’ and ‘large overshoots/undershoots (spikes)’ are solved

3 | DISCUSSION

The goal of this research is not to present novel circuits for SRAM-bitcell, but is to investigate the challenges and solutions of working at ‘ $V_{DD} = V_{th}$ ’ and also to present unreported ability of power-gating technique in adjusting V_{DD} on V_{th} without any instabilities in bitcell performance, which these

TABLE 3 Performance summary of the proposed sram bitcells

Work	Technology	Write Delay (s)	Read Delay (s)	Leakage Current (pA)	Static Power (pW)	Dynamic Power (nW)	Write SNM (mV)	Hold SNM (mV)	Read SNM (mV)	V _{DD} (V)	$\frac{SNM_{min}}{V_{DD}} \times 100 = SNM\%$
11T Cell (Figure 9)	32 nm CNTFET	6.11 n	3.27 n	218.1	65.43	1.21	156.3	156.2	156.2	0.3	$\frac{156.2}{300} \times 100 = 52\%$
8T Cell (Figure 5)	32 nm CNTFET	6.11 n	119 p	54.82	16.45	0.31	156.3	130.5	156.4	0.3	$\frac{130.5}{300} \times 100 = 43\%$
6T Cell (Figure 2)	32 nm CNTFET	69.9 p	156 p	272.6	89.97	9.29	143.1	140.1	147.3	0.33	$\frac{140.1}{330} \times 100 = 42\%$

solutions lead to having an optimum bitcell. To this end, first of all, the standard 6T bitcell is regenerated to present the challenges: ‘read-mode discharges (distortions)’, ‘large overshoots/undershoots (spikes)’ and ‘drop of Logic-1 voltage-level in read-mode’. Note that although other CNTFET works referred to similar problems [10–75], all of them were done at $V_{DD} > V_{th}$, hence they did not face the challenges of working at threshold supply-voltage. It is obvious that working at ‘ $V_{DD} > V_{th}$ ’ will lead to lower power-consumption (Tables 1–3).

The regenerated results at $V_{DD} > V_{th}$ reveal large undershoots (~ 500 nA, -1μ A) in the start-up current in Figure 3e, so Logic-1 voltage-level drops sharply below V_{th} in read mode, and the CNTFETs cannot turn on/off perfectly (see Figures 3c and d) that it causes a failure in the bitcell operation. Regenerating at nearthreshold supply-voltage ($V_{DD} > V_{th}$) solves those drops though ‘read-mode discharges (distortions)’ and ‘larger undershoots/overshoots ($\pm 2\mu$ A)’ still occur, see Figures 4d and e.

By applying power-gating technique, the 8T bitcell exhibits small overshoots and undershoots (± 50 nA) under threshold supply-voltage ($V_{DD} = V_{th}$), see Figure 6e. Actually, this study reveals the ability of power-gating technique in adjusting V_{DD} on V_{th} without any instabilities that this feature was not reported before [41, 42, 63–65]. Albeit this 8T bitcell solves ‘sharp overshoots/undershoots (spikes)’, but it cannot make any solutions for ‘read-mode discharges (distortions)’ and ‘the drop of Logic-1 voltage-level in read mode’, see Figure 6d.

Finally, a power-gating 11T SRAM-bitcell composed of ‘decoupled R/W BLs’ and ‘an output inverter in read path’ (Figure 9) succeeds to solve all challenges (Figure 10). It can keep the spikes at ± 50 nA (Figure 10e) and render a fine subthreshold performance at nodes Q/QN (Figure 10c) at $V_{DD} = V_{th}$ without any problems in read mode (Figure 10d), while it causes ‘drop of Logic-1 voltage-level in read mode’ in the 6T and 8T ones (Figures 4d and 6d). In fact, this feature of power-gating and also the idea of employing output inverter in read path, lead to have a stable bitcell with small powers, compared to other works (Tables 1–3). Other works often reduced delays at the cost of higher power-consumptions [10–75].

Although this work is not experimental but it applies the Monte Carlo method to demonstrate the reliability of this study. Figures 11–14 estimate the function of 6T, 8T and 11T bitcells over V_{DD} . Even though the leakage current and static/dynamic powers of 11T bitcell are higher than 8T one (Figures 12–14), it has better V_{data} (Figure 11) and succeeds to solve the challenges. Also, its leakage-current and powers are still smaller than ‘all 6T bitcells in Table 1’ and ‘other 8T bitcells in Table 2’. Maybe it seems that using 11 Transistors causes some foot-print problems, but note that other works utilised 10, 11, 16, 18, 20 and even 22 Transistors in their bitcells without any problems [12, 13, 15, 19, 20, 23, 28, 33, 58, 68]. As a result, the 11T design in Figure 9 is the optimum SRAM-bitcell that this study tries to obtain it by solving the challenges. It clearly illustrates the challenges, solutions, and unreported ability of power-gating technique.

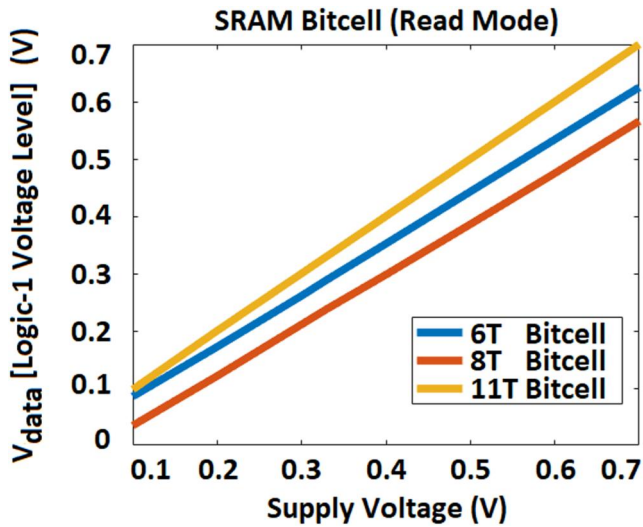


FIGURE 11 Changes of Logic-1 voltage-level in read mode (V_{data}) over supply voltage (V_{DD}), for the proposed 6T, 8T and 11T Static Random Access Memory (SRAM) bitcells. Note that, as illustrated in Figures 3 and 4, the 6T bitcell does not work at $V_{DD} = 0.3$ V and its practical area is $V_{DD} \geq 0.33$ V

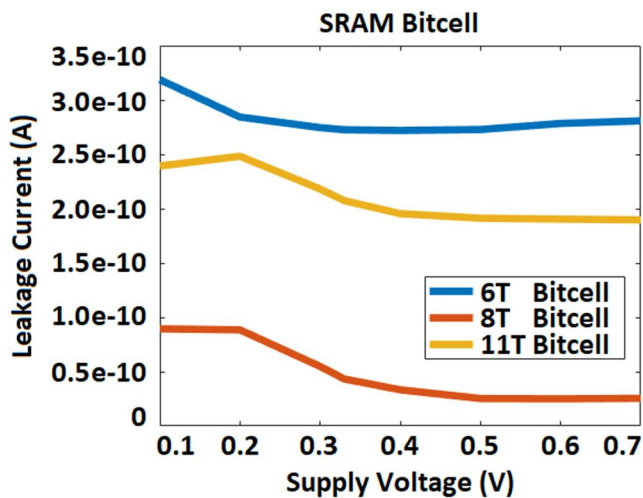


FIGURE 12 Changes of leakage current (I_{leak}) over supply voltage (V_{DD}), for the proposed 6T, 8T and 11T Static Random Access Memory (SRAM) bitcells. Note that, the 6T SRAM bitcell does not work at $V_{DD} = 0.3$ V because its true practical region is $V_{DD} \geq 0.33$ V

Although Adam Teman and Alexander Fish in 2010 rendered a similar work in the Sub-threshold and Near-threshold region [84], their work had major differences as follow:

- (1) They considered the challenges ahead by just reviewing the previous MOSFET works in 2007, 2008 and 2010 which tried to solve the related challenges but still had microwatt power dissipations; while this paper is investigating and solving them at Nano and Pico watt power-consumptions.
- (2) They referred to the Subthreshold supply-voltage works (200–350mV) in 2007, 2008 and 2010; while this paper is proposing an optimum 11T Threshold supply-voltage

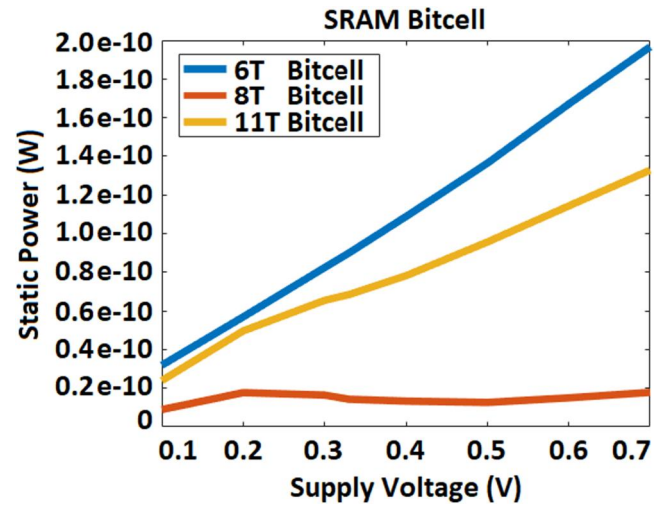


FIGURE 13 Changes of static power (P_s) over supply voltage (V_{DD}), for the proposed 6T, 8T and 11T Static Random Access Memory (SRAM) bitcells. Figures 3 and 4 illustrate that the practical area of 6T bitcell is $V_{DD} \geq 0.33$ V and it does not work at $V_{DD} = 0.3$ V

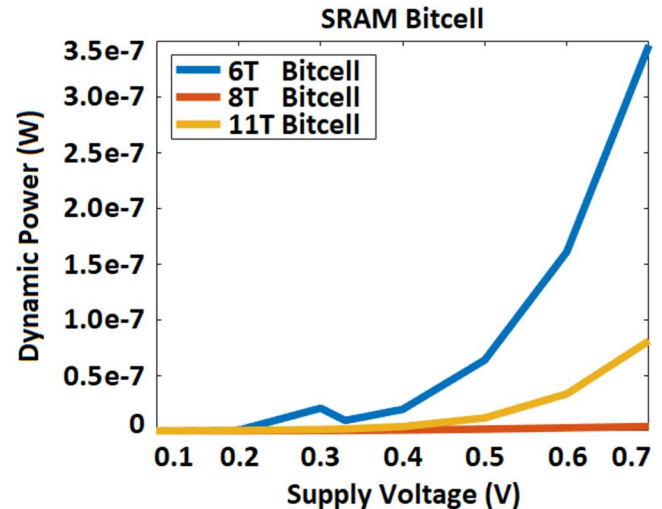


FIGURE 14 Changes of dynamic power (P_D) over supply voltage (V_{DD}), for the proposed 6T, 8T and 11T Static Random Access Memory (SRAM) bitcells. As mentioned before, the 6T SRAM bitcell does not work at $V_{DD} = 0.3$ V because its practical region is $V_{DD} \geq 0.33$ V

work (300mV) with Subthreshold operations at Q and QB nodes (around 200mV) which has 300mV output as Logic 1 in read mode and is able to solve the leakage current, read mode and also SNM problems. Also, its SNM % is 52.

- (3) This paper presents the new feature of power-gating technique which was not reported in Ref. [84]. It also illustrates the problem of using separate read path (Figures 7, 8) which the read access-switch is after the inverter, but Ref. [84] did not mention anything about this matter in the 2007, 2008 and 2010 works.
- (4) This paper introduces SNM% as a useful figure-of-merit in comparing different works with various supply-voltages.

4 | CONCLUSION

As mentioned by S. R. Prasad et al in 2012, albeit employing nanotechnologies leads to the reduction of V_{DD} and dynamic-power, transistor-scaling causes a considerable increase in leakage current, as a result, digital circuits such as SRAMs have been impressed strongly. Thanks to carbon nanotubes, CNTFET as a promising candidate for CMOS and mighty competitor for FinFET, along with effective techniques such as power-gating, has covered the defects of nanotechnology during recent years. This study investigates the possibility, challenges and solutions of working at $V_{DD} = V_{th}$, because working at threshold supply-voltage will lead to lower power-consumptions. In this respect, the standard 6T bitcell is regenerated and compared in order to reveal the challenges ahead. Then a power-gating 8T bitcell is analysed to discuss an unreported feature of power-gating technique. Finally, an optimum 11T bitcell without any instabilities or voltage-reductions is rendered at threshold supply-voltage while it has fine sub-threshold operation at nodes Q and QN. This 11 T bitcell with the leakage current of 218.1 pA, static power of 65.43 pW, dynamic power of 1.21 nW, and the 52% SNM succeeds to solve the three challenges: 'drop of Logic-1 voltage-level in read mode', 'large overshoots/undershoots (spikes)' and 'read-mode discharges (distortions)'. This 11 T bitcell is designed based on the solutions explained during the article. It shows the unreported feature of power-gating. Investigation of Temperature effects on working under Threshold Supply-Voltage can be the future research.

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CONFLICT OF INTEREST

The author declares that there is no conflict of interest associated in this manuscript.

DATA AVAILABILITY STATEMENT

Data sharing not applicable—no new data generated, or the article describes entirely theoretical research.

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REFERENCES

- Norman, R.: Solid State Switching and Memory Apparatus. U. S. Patent 3562721 Filed Mar. 5, 1963, Issued Feb. 9, 1971) <https://patents.google.com/patent/US3562721>
- Laws, D.A.: A company of legend: the legacy of fairchild semiconductor. *IEEE Ann. Hist. Comput.* 32(1), 60–74 (2010). <https://doi.org/10.1109/MAHC.2010.12>
- Prasad, S.R., Madhavi, B.K., Kishore, K.L.: "Design of low-leakage CNTFET SRAM cell at 32nm technology using Forced Stack technique. *Int Eng Res App.* 2(1), 805–808 (2012)
- Morifuji, E., et al.: New guideline of Vdd and Vth scaling for 65 nm technology and beyond. In: Symposium on VLSI Technology Digest of Technical Papers, pp. 164–165 (2004). <https://doi.org/10.1109/VLSIT.2004.1345457>
- Iijima, S.: Helical microtubules of graphitic carbon. *Nature* 354(6348), 56–58 (1991). <https://doi.org/10.1038/354056a0>
- Iijima, S., Ichihashi, T.: Signal-shell carbon nanotubes of 1-nm diameter. *Nature* 363(6430), 603–605 (1993). <https://doi.org/10.1038/363603a0>
- Bethune, D.S., et al.: Cobalt-catalyzed growth of carbon nanotubes with single-atomic-layer walls. *Nature* 363(6430), 605–607 (1993). <https://doi.org/10.1038/363605a0>
- Dekker, C., Tans, S.J., Verschueren, A.R.M.: Room-temperature transistor based on a single carbon nanotube. *Nature* 393(6680), 49–52 (1998). <https://doi.org/10.1038/29954>
- Martel, R., et al.: Single- and multi-wall carbon nanotube field-effect transistors. *Appl. Phys. Lett.* 73(17), 2447–2449 (1998). <https://doi.org/10.1063/1.122477>
- Emon, D.H., Mohammad, N., Mominuzzaman, M.: Design of a low standby power CNFET based SRAM Cell. In: 7th International Conference on Electrical and Computer Engineering (2012). <https://doi.org/10.1109/ICECE.2012.6471523>
- Kanhaiya, P.S., et al.: 1 kbit 6T SRAM arrays in carbon nanotube FET CMOS. In: Symposium on VLSI Technology Digest of Technical Papers (2019). <https://doi.org/10.23919/VLSIT.2019.8776563>
- Prasad, S.R., Madhavi, B.K., Kishore, K.L.: Design of 32nm forced stack CNTFET SRAM cell for leakage power reduction. In: International Conference on Computing, Electronics and Electrical Technologies (ICCEET) (2012). <https://doi.org/10.1109/ICCEET.2012.6203904>
- Murotiya, S.L., Gupta, A.: CNTFET based design of content addressable memory cells. In: 4th International Conference on Computer and Communication Technology (ICCT) (2013). <https://doi.org/10.1109/ICCT.2013.6749593>
- Singh, A., Khosla, M., Raj, B.: CNTFET modeling and low power SRAM cCell design. In: IEEE 5th Global Conference on Consumer Electronics (2016). <https://doi.org/10.1109/GCCE.2016.7800437>
- Srinivasan, P., Bhat, A.S., Murotiya, S.L.: Anu gupta, "design and performance evaluation of a low transistor Ternary CNTFET SRAM Cell. In: International Conference on Electronic Design, Computer Networks & Automated Verification (EDCAV) (2015). <https://doi.org/10.1109/EDCAV.2015.7060535>
- Niranjan, S., Shanmukha, S., Vasundara, P.K.S.: A novel design of ternary level SRAM cell using CNTFET. In: International Conference on Networking, Embedded and Wireless Systems (ICNEWS) (2018). <https://doi.org/10.1109/ICNEWS.2018.8904018>
- Elangovan, M., Gunavathi, K.: Stability analysis of 6T CNTFET SRAM cell for single and multiple CNTs. In: 4th International Conference on Devices, Circuits and Systems (ICDCS) (2018). <https://doi.org/10.1109/ICDCSyst.2018.8605154>
- Spasova, M., et al.: SRAM design based on carbon nanotube Field effect transistor's model with modified Parameters. In: 40th International Spring Seminar on Electronics Technology (ISSE) (2017). <https://doi.org/10.1109/ISSE.2017.8000953>
- Das, S., Das, D., Rahaman, H.: Design of content addressable memory cell using carbon nanotube Field effect transistors. In: IEEE Students' Technology Symposium (2016). <https://doi.org/10.1109/TechSym.2016.7872669>
- Kanhaiya, P.S., et al.: Carbon Nanotube-based CMOS SRAM: 1 kbit 6T SRAM arrays and 10T SRAM Cells. *IEEE Trans. Electron. Dev.* 66(12), 5375–5380 (2019). <https://doi.org/10.1109/TED.2019.2945533>
- Prasad, S.R., Madhavi, B.K., Kishore, K.L.: Design of a 32nm 7T SRAM cell based on CNTFET for low power operation. In: International Conference on Devices, Circuits and Systems (ICDCS) (2012). <https://doi.org/10.1109/ICDCSyst.2012.6188756>
- Pushkama, A., Raghavan, S., Mahmoodi, H.: Comparison of performance parameters of SRAM designs in 16nm CMOS and CNTFET technologies. In: 23rd IEEE International SOC Conference (2010). <https://doi.org/10.1109/SOCC.2010.5784690>

23. Liu, Z., et al.: Design of a novel ternary sram sense amplifier using CNFET. In: IEEE 12th International Conference on ASIC (ASICON) (2017). <https://doi.org/10.1109/ASICON.2017.8252448>
24. Shahdipour, H., et al.: Effects of CNT diameter variability on a CNFET-based SRAM. In: IEEE Asia Pacific Conference on Circuits and Systems (2010). <https://doi.org/10.1109/APCCAS.2010.5774837>
25. Mohita, T.N., et al.: Design and stability analysis of CNTFET based SRAM cell. In: Students' Conference on Electrical, Electronics and Computer Science (SCEECS) (2016). <https://doi.org/10.1109/SCEECS.2016.7509327>
26. Bala, Sh., Khosla, M.: Design and performance analysis of low-power SRAM based on Electrostatically doped Tunnel CNTFETs. *J. Comput. Electron.* 18(3), 856–863 (2019). <https://doi.org/10.1007/s10825-019-01345-z>
27. Jadon, A., Akashe, Sh.: Memristive power optimization of non-volatile seven transistors static random access memory Cell. In: Advances in Optical Science and Engineering, Part of the Springer Proceedings in Physics Book Series (SPPHY, vol. 166 (2015). https://doi.org/10.1007/978-81-322-2367-2_31
28. Elangovan, M., Gunavathi, K.: High stability and low-power dual supply-stacked cntfet sram cell. In: Innovations in Electronics and Communication Engineering, Lecture Notes in Networks and Systems Book Series (LNNS), vol. 33 (2018). https://doi.org/10.1007/978-981-10-8204-7_21
29. Amdapurkar, A., et al.: A modified GDI-based low-power and high read stability 8-T SRAM memory with CNTFET technology. In: Nano-electronic Materials and Devices, Part of the Lecture Notes in Electrical Engineering Book Series (LNEE), vol. 466 (2017). https://doi.org/10.1007/978-981-10-7191-1_15
30. Kumar, G.S., Singh, A., Raj, B.: Design and analysis of a gate-all-around CNTFET-based SRAM cell. *J. Comput. Electron.* 17(Sep), 138–145 (2017). <https://doi.org/10.1007/s10825-017-1056-x>
31. Prasad, S.R., Madhavi, B.K., Kishore, K.L.: Low leakage-power SRAM cell design using CNTFETs at 32nm technology. In: Advances in Communication, Network, and Computing, Part of the Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering Book Series (LNICST), vol. 108 (2012). https://doi.org/10.1007/978-3-642-35615-5_24
32. Prasad, S.R., Madhavi, B.K., Kishore, K.L.: Data-retention sleep transistor CNTFET SRAM cell design at 32nm technology for low-leakage. In: Mobile Communication and Power Engineering, Part of the Communications in Computer and Information Science Book Series (CCIS), vol. 296 (2012). https://doi.org/10.1007/978-3-642-35864-7_54
33. Satyanarayana, S.V.V., et al.: Dual-chirality GAA-CNTFET-based SCPF-TCAM cell design for low power and high performance. *J. Comput. Electron.* 18(Jun), 1045–1054 (2019). <https://doi.org/10.1007/s10825-019-01362-y>
34. Zhang, Zh., Delgado-Frias, J.G.: Low power and metallic CNT tolerant CNTFET SRAM design. In: 11th IEEE International Conference on Nanotechnology Portland Marriott (2011). <https://doi.org/10.1109/NANO.2011.6144323>
35. Zhang, Zh., Delgado-Frias, J.G., Nyathi, J.: CNTFET SRAM cell design with tolerance to metallic CNTs. In: 53rd IEEE International Midwest Symposium on Circuits and Systems (2010). <https://doi.org/10.1109/MWSCAS.2010.5548846>
36. Kim, Y., et al.: Novel 8-T CNFET SRAM cell design for the future ultra-low power microelectronics. In: International SoC Design Conference (ISODC) (2016). <https://doi.org/10.1109/ISODC.2016.7799768>
37. Zhang, Zh., Delgado-Frias, J.G.: CNTFET SRAM cell with tolerance to removed metallic CNTs. In: IEEE 55th International Midwest Symposium on Circuits and Systems (MWSCAS)Aug (2012). <https://doi.org/10.1109/MWSCAS.2012.6291988>
38. Zhang, Zh., et al.: Performance of CNFET SRAM cells under diameter variation corners. In: 52nd IEEE International Midwest Symposium on Circuits and Systems (2009). <https://doi.org/10.1109/MWSCAS.2009.5236035>
39. Zhang, Zh., Delgado-Frias, J.G.: CNTFET 8T SRAM cell performance with near-threshold power supply scaling. In: IEEE International Symposium on Circuits and Systems (ISCAS) (2013). <https://doi.org/10.1109/ISCAS.2013.6572293>
40. Zhang, Zh., Delgado-Frias, J.G.: Near-threshold CNTFET SRAM cell design with word-line Boosting and removed metallic CNT Tolerance. *IEEE Trans. Nanotechnol.* 13(2), 182–191 (2014). <https://doi.org/10.1109/TNANO.2013.2295757>
41. Zhang, Zh., Delgado-Frias, J.G.: Near-threshold CNTFET SRAM cell design with gated cell power supply. In: IEEE 56th International Midwest Symposium on Circuits and Systems (MWSCAS)Aug (2013). <https://doi.org/10.1109/MWSCAS.2013.6674655>
42. Delgado-Frias, J.G., Zhang, Zh., Turi, M.A.: Near-threshold CNTFET SRAM cell design with removed metallic CNT Tolerance. In: IEEE International Symposium on Circuits and Systems (ISCAS) (2015). <https://doi.org/10.1109/ISCAS.2015.7169300>
43. Sun, L., et al.: A low power and Robust carbon nanotube 6T SRAM design with metallic tolerance. In: Design, Automation & Test in Europe Conference & Exhibition (DATE) (2014). <https://doi.org/10.7873/DATE.2014.125>
44. Zhang, Zh., Delgado-Frias, J.G.: Carbon nanotube SRAM design with metallic CNT or removed metallic CNT Tolerant Approaches. *IEEE Trans. Nanotechnol.* 11(4), 788–798 (2012). <https://doi.org/10.1109/TNANO.2012.2197636>
45. Delgado-Frias, J.G., Zhang, Zh., Turi, M.A.: Low power SRAM cell design for FinFET and CNTFET technologies. In: International Conference on Green Computing (2010). <https://doi.org/10.1109/GREENCOMP.2010.5598266>
46. Patel, P.K., Malik, M.M., Gupta, T.: Optimization techniques for high performance 9T SRAM cell design. In: First International Conference on Information and Communication Technology for Intelligent Systems, vol. 1 (2016). https://doi.org/10.1007/978-3-319-30933-0_28
47. Patel, P.K., Malik, M.M., Gupta, T.K.: Reliable high-yield CNTFET-based 9T SRAM operating near threshold voltage region. *J. Comput. Electron.* 17, 774–783 (2018). <https://doi.org/10.1007/s10825-017-1127-z>
48. Moradinassab, M., Karbassian, F., Fathipour, M.: A comparison study of the effects of supply voltage and Temperature on the stability and performance of CNFET and nanoscale Si-MOSFET SRAMs. In: 1st Asia Symposium on Quality Electronic Design (2009). <https://doi.org/10.1109/ASQED.2009.5206304>
49. Dhillewarara, P., Mahapatra, R., Srinivas, P.S.T.N.: High SNM 32nm CNFET based 6T SRAM cell design considering transistor ratio. In: International Conference on Electronics and Communication System (ICECS) (2014). <https://doi.org/10.1109/ECS.2014.6892748>
50. Ahmed, Z., et al.: Low voltage SRAM design using tunneling regime of CNTFET. In: 14th IEEE International Conference on Nanotechnology (2014). <https://doi.org/10.1109/NANO.2014.6968042>
51. Lin, Sh., Kim, Y.-B., Lombardi, F.: Design of a CNTFET-based SRAM cell by dual-chirality selection. *IEEE Trans. Nanotechnol.* 9(1), 30–37 (2010). <https://doi.org/10.1109/TNANO.2009.2025128>
52. Ramakrishna, S.B., Aswatha, A.R.: Device characterization and impact of CNT diameter variation on the read write stability of CNTFET 6t SRAM cell. In: International Conference on Innovative Mechanisms for Industry Applications (ICIMIA) (2017). <https://doi.org/10.1109/ICIMIA.2017.7975523>
53. Arif, Sh., Pal, S.: CNFET based 8-T SRAM cell to improve stability with reduced power consumption. In: International Conference on Advanced Computing and Communication Systems (ICACCS) (2015). <https://doi.org/10.1109/ICACCS.2015.7324112>
54. Wang, W., Yu, Zh., Choi, K.: High SNM 6T CNFET SRAM cell design considering nanotube diameter and transistor ratio. In: IEEE International Conference on Electro/Information Technology (2011). <https://doi.org/10.1109/EIT.2011.5978618>
55. Kim, Y.B., et al.: A low power 8T SRAM cell design technique for CNFET. In: International SoC Design Conference (2008). <https://doi.org/10.1109/SOCDC.2008.4815601>

56. Islam, A., Hasan, M.: Power and variability aware design of SRAM using carbon nanotube Field effect transistor. In: Annual IEEE India Conference (INDICON) (2010). <https://doi.org/10.1109/INDCON.2010.5712597>
57. Moradinassab, M., Fathipour, M.: Stable, low power and high performance SRAM based on CNFET. In: 10th International Conference on Ultimate Integration of Silicon (2019). <https://doi.org/10.1109/ULIS.2009.4897599>
58. Shreya, S., Sourav, S.: Design, analysis and comparison between CNTFET based ternary SRAM cell and PCRAM Cell. In: International Conference on Communication, Control and Intelligent Systems (CCIS) (2015). <https://doi.org/10.1109/CCIS.2015.7437938>
59. Lin, Sh., et al.: A new SRAM cell design using CNTFETs. In: International SoC Design Conference (2008). <https://doi.org/10.1109/SOCD.2008.4815599>
60. Singh, P., Chandel, R., Sharma, N.: Stability analysis of SRAM cell using CNT and GNR Field effect transistors. In: Tenth International Conference on Contemporary Computing (IC3) (2017). <https://doi.org/10.1109/IC3.2017.8284335>
61. Arif, Sh., Pal, S.: Variation-resilient CNFET-based 8T SRAM cell for ultra-low-power application. In: International Conference on Signal Processing and Communication Engineering Systems (2015). <https://doi.org/10.1109/SPACES.2015.7058235>
62. Shahrabadi, S.: Investigating PDP-based error by testing NAND and D-Latch based on CNFET (Stanford model). *SN Appl. Sci.* 1(7), 702 (2019). <https://doi.org/10.1007/s42452-019-0742-7>
63. Turi, M.A., Delgado-Frias, J.G.: Effective low leakage 6T and 8T FinFET SRAMs: using cells with reverse-biased FinFETs, near-threshold operation, and power gating. *IEEE Trans Circ Sys: Expr Brif* 67(4), 765–769 (2020). <https://doi.org/10.1109/TCSII.2019.2922921>
64. Liu, K., et al.: A 373-F^2 0.21%-native-BER EE SRAM physically unclonable function with 2-D power-Gated bit cells and V_{SS} bias-based dark-bit detection. *IEEE J. Solid State Circ.* 55(6), 1–14 (2020). <https://doi.org/10.1109/JSSC.2019.2963002>
65. You, W.-X., Su, P., Hu, Ch.: A new 8t hybrid nonvolatile sram with ferroelectric FET. *IEEE J Elec Dev Soci* 8, 171–175 (2020). <https://doi.org/10.1109/JEDS.2020.2972319>
66. Patel, P.K., Malik, M., Gupta, T.K.: Design of an Ultralow power CNTFET based 9T SRAM with shared BL and Half select free techniques. *Int. J. Numer. Model. Electron. Network. Dev. Field.* 32(2), e2511 (2019). <https://doi.org/10.1002/jnm.2511>
67. Kumar, M., Ubhi, J.S.: Design and analysis of CNTFET based 10T SRAM for high performance at nanoscale. *Int. J. Circ. Theor. Appl.* 47(11), 1775–1785 (2019). <https://doi.org/10.1002/cta.2696>
68. Shrivastava, Y., Gupta, T.K.: Design of high-speed low variation static Noise Margin ternary S-RAM cells. *IEEE Trans. Device Mater. Reliab.* 21(1), 102–110 (2021). <https://doi.org/10.1109/TDMR.2021.3058159>
69. Adib, N.S., et al.: Performance analysis of CNTFET based 6T SRAM cell with Tube and chiral variation under 16 and 32 nm Technologies. In: 6th International Conference for Convergence in Technology (I2CT) (2021). <https://doi.org/10.1109/I2CT51068.2021.9417988>
70. Kim, D., et al.: An SRAM compiler for monolithic-3-d integrated circuit with carbon nanotube transistors. *IEEE J Expl Solid-State Com Devi Circuits* 7(2), 106–114 (2021). <https://doi.org/10.1109/JXDC.2021.3120715>
71. Chen, R., et al.: Carbon nanotube SRAM in 5-nm technology node design, optimization, and performance evaluation—part i: cnfet transistor optimization. *IEEE Trans. Very Large Scale Integr. Syst.* 30(4), 432–439 (2022). <https://doi.org/10.1109/TVLSI.2022.3146125>
72. Ghodrati, M., Mir, A., Farmani, A.: Non-destructive label-free biomaterials detection using tunneling carbon nanotube-based biosensor. *IEEE Sensor. J.* 21(7), 8847–8854 (2021). <https://doi.org/10.1109/JSEN.2021.3054120>
73. Ghodrati, M., Farmani, A., Mir, A.: Nanoscale sensor-based tunneling carbon nanotube transistor for toxic gases detection: a first-principle study. *IEEE Sensor. J.* 19(17), 7373–7377 (2019). <https://doi.org/10.1109/JSEN.2019.2916850>
74. Khaleqi Qaleh Jooqa, M., et al.: A robust and energy-efficient near-threshold sram cell utilizing ballistic carbon nanotube wrap-gate transistors. *Int J Elect Commun* 110, 152874 (2019). <https://doi.org/10.1016/j.aecue.2019.152874>
75. Khaleqi Qaleh Jooqa, M., et al.: Semi-analytical modeling of high performance nano-scale complementary logic gates utilizing Ballistic carbon nanotube transistors. *Phys. E Low-dimens. Syst. Nanostruct.* 104, 286–296 (2018). <https://doi.org/10.1016/j.physe.2018.08.008>
76. Almudever, C.G., Rubio, A.: Variability and reliability analysis of CNFET technology: impact of manufacturing imperfections. *Microelectron. Reliab.* 55(2), 358–366 (2015). <https://doi.org/10.1016/j.microrel.2014.11.011>
77. Mitra, S., et al.: Imperfection-immune VLSI logic circuits using carbon nanotube Field effect transistors. In: Design, Automation & Test in Europe Conference & Exhibition (2009). <https://doi.org/10.1109/DATE.2009.5090705>
78. Amat, E., et al.: Systematic and random variability analysis of two different 6T-SRAM layout topologies. *Microelectron. J.* 44(9), 787–793 (2013). <https://doi.org/10.1016/j.mejo.2013.06.010>
79. Pouyan, P., Amat, E., Rubio, A.: adaptive proactive reconfiguration: a technique for process-variability- and aging-aware sram cache design. *IEEE Trans. Very Large Scale Integr. Syst.* 23(9), 1951–1955 (2015). <https://doi.org/10.1109/TVLSI.2014.2355873>
80. Shahrabadi, S.: Ultrawideband LNA 1960–2019: review. *IET Circuits, Devices Syst.* 15(8), 697–727 (2021). <https://doi.org/10.1049/cds.2.12071>
81. Shahrabadi, S.: Noise figure and input intercept point's errors in an ageless microwatt ultrawideband system (limiter and low noise amplifier). *IET Commun.* 15(20), 2597–2614 (2021). <https://doi.org/10.1049/cmu.2.12308>
82. Shahrabadi, S., Moravvej-Farshi, M.K.: Investigation of a nano-scale compound metamaterial for the design of Optical Filters. *Electron. World* 121(1949) (2015). <http://viewer.zmags.com/publication/5a3df92c%23/5a3df92c/1>
83. Bol, D., Flandre, D., Legat, J.-D.: Nanometer MOSFET effects on the minimum-energy point of sub-45nm subthreshold logic mitigation at technology and circuit levels. *ACM Trans. Des. Autom. Electron. Syst.* 16(1)26 (2010). <https://doi.org/10.1145/1870109.1870111>
84. Teman, A., Fish, A.: Sub-threshold and near-threshold SRAM design. In: IEEE 26-th Convention of Electrical and Electronics Engineers in Israel (2010). <https://doi.org/10.1109/EEEL.2010.5662147>

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